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Part 4: Reflection

The Firestore transaction for stock adjustments took longer than I initially expected. A basic implementation using `updateDoc` was straightforward, but ensuring that both the stock quantity update and the inventory movement record were written atomically required using `runTransaction`. Learning how Firestore transactions work, including automatic retries during conflicts, was one of the more valuable parts of this assignment.

If I were to start the project again, I would implement the Firestore security rules before developing the application features. I initially built the application using open test-mode rules and configured authentication restrictions only at the end. As a result, I had to re-test every database operation after enabling authentication, which could have been avoided with a security-first approach.

The area I am least confident in is designing scalable Firestore data models. In particular, I would like to gain a better understanding of when to denormalize data, such as storing product information directly in inventory movement records, versus keeping a normalized structure and retrieving related data from separate documents. Developing a stronger understanding of NoSQL data modelling is an area I plan to continue improving.